

Contact :

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Employment History :

Jul. 2020 - Current: Department of Physics, IIT Hyderabad.
Assistant Professor

Jun. 2015 - Jun. 2020: Department of Physics, University of Cincinnati, USA
Post-Doctoral Fellow based at KEK, Tsukuba, Japan

Education :

Doctorate
(2011 - 2015) Tata Institute of Fundamental Research, Mumbai.
Thesis title : 'Study of Bottomonium states in Radiative $\Upsilon(2S)$ decays at Belle'.

Pre-PhD
(2009 - 2011) Experimental projects on 'Characterization of Silicon Micro-Strip Detectors' and 'Characterization of Silicon Photo-Multipliers'.

M.Sc. (Physics)
(2007 - 2009) Banaras Hindu University, Varanasi.
Special courses on "Nuclear and Particle Physics", a dissertation on 'Hawking Radiation'

B.Sc. (honors) Physics
(2004 - 2007) Banaras Hindu University, Varanasi.

Research Interest/Experience :

- Physics analyses of $e^+ e^-$ collision data recorded by Belle (II) experiment.
 - Rare B meson decays.
 - Hadronic decays of B-mesons.
 - Bottomonium states and their transitions.
- Detector Development for Belle II.
 - Leading contribution in the fabrication of the novel imaging time-of-propagation (TOP) counters and significant contribution in its commissioning.
 - Charged kaon and pion identification performance studies for the Belle II Detector focused on TOP
 - Participated in the assembly of the layer-4 of its Silicon Vertex Detector.
- Service to the Belle II Collaboration:
 - Convener (Jan, 2018 - Jan, 2022) and Data Production Contact (Feb, 2017 - May, 2019) for the Electro-Weak Penguin physics working group in Belle II Collaboration.
 - Coordinator (Jul 2019 - Oct 2020) for the calibration efforts of the TOP detector.

Selected Recent Publications :

- “Measurement of $B \rightarrow K^*(892)\gamma$ decays at Belle II”, Belle II Collaboration, arXiv:2411.10127 (accepted for publication Journal of High Energy Physics)
- “The imaging Time-of-Propagation detector at Belle II”, H. Atmacan, M. Belhorn, Y. Guan, L. Li, B. Pal, S. Sandilya et. al, Nuclear Inst. and Methods in Physics Research, A 1080 (2025) 170627.
- "Charged-hadron identification at Belle II", Belle II Collaboration, arXiv:2506.04355 (European Physical Journal C)
- “A new Monte Carlo generator for BSM physics in $B \rightarrow K^*\ell^+\ell^-$ decays with an application to lepton non-universality in angular distributions”, Alexei Sibidanov, Thomas E. Browder, Shawn Dubey, Shahab Kohani, Rusa Mandal, Saurabh Sandilya, Rahul Sinha and Sven E. Vahsen, JHEP 08 (2024) 151.
- “Search for Rare $b \rightarrow d\ell^+\ell^-$ transitions at Belle,” I. Adachi et al. [Belle and Belle II Collaboration (includes S. Sandilya)], Phys. Rev. Lett. 133, no.10, 101804 (2024),
- “Measurement of the B^+/B^0 production ratio in e^+e^- collisions at the $\Upsilon(4S)$ resonance using $B \rightarrow J/\psi(\ell\ell)K$ ”, Belle Collaboration, Phys. Rev. D 107, no.3, L031102 (2023).
- “The design, construction, operation, and performance of the Belle II silicon vertex detector”, Belle II SVD group, JINST 17, no.11, P11042 (2022).
- A complete list of publications: <https://inspirehep.net/authors/1223031>

Selected Recent Talks:

- Invited departmental talk on “Charged particle identification in Belle II (and LFU measurement)” at Tokyo City University (Jul 25th, 2023).
- Parallel talk on ‘Charged particle identification performance of the TOP counters in Belle II’ at International Conference on Technology and Instrumentation in Particle Physics 2021, TRIUMF, Vancouver, Canada (May 24 - 28, 2021).
- Plenary talk on ‘Latest Belle II results on beauty and charm decays’ at the 55th Rencontres de Moriond - QCD & High Energy Interactions (Moriond QCD 2021) conference, (Mar 27 - Apr 3, 2021) La Thuile, Italy.
- Plenary Review Talk on ‘Lepton Flavor Universality tests in semileptonic B decays’ at the (FPCP 2020) conference, (Jun 08-12, 2020) A Toxa, Spain.
- Plenary Talk on ‘Rare Decays and Lepton Flavor Universality’ on behalf of the Belle (II) and LHCb Collaborations at the 2019 Meeting of the American Physics Society Division of Particles and Fields (DPF 2019), (Jul 29-Aug 02, 2019) Northeastern University, Boston, USA.
- Plenary Talk on ‘ $b \rightarrow s\ell\ell$ transitions at Belle & Belle II’ on behalf of the Belle (II) collaborations at the ‘6th KEK Flavor Factory Workshop (KEKFF 2019)’, (Feb 14-16, 2019), KEK, Tsukuba, Japan.

Teaching Activities:

- Teaching courses such as ‘Particle Physics’, ‘Radiation Detection, Instrumentation, and Measurement’ and laboratory-related courses for undergraduate and postgraduate students at IIT Hyderabad.

- Started and co-ordinating MSc in Medical Physics at IIT Hyderabad.
- Invited lecture on the occasion of National Science Day (Feb 28th, 2023) on “High Energy Physics: A Global Science” at Central University of Karnataka.
- Lecture on ‘TOP and Particle Identification’ at the 5th Belle II Summer School, at Virginia Tech., Virginia, USA. (Jul 12-17, 2021)
- Lecture on ‘Meet the Anomalies: Experimental Perspective on $b \rightarrow s\ell\ell$ ’ at first advanced school on flavor physics, at LMU Munich, Germany (Mar 22-26, 2021).
- Lecture on ‘Charged Particle Identification at Belle II’ in Belle Analysis Workshop 2017 at MNIT, Jaipur.
- Tutorial on ‘Monte-Carlo event generation in Belle framework’ in Belle II Analysis Software Tutorial 2016, (Jan-2016) KEK, Japan.
- Conducted tutorial sessions in fitting data using RooFit during Belle Analysis Workshops 2013 and 2014 held at TIFR and IIT Guwahati, respectively.
- Talks/Lectures in Belle Analysis Workshop (BAW), (Jan-2013 and Jan-2012) TIFR, India.
- Tutored a course on ‘Experimental High Energy Physics’ at TIFR, Mumbai.
- Qualified for joint CSIR-UGC test for eligibility for Lectureship.

Service to HEP Community:

- Convener of “Rare B, D and K decays, radiative and electroweak penguin decays, including constraints on V_{td}/V_{ts} and ϵ'/ϵ ” working group in 12th International Workshop on the CKM Unitarity Triangle (2023) in Santiago de Compostela, Spain.
- Convener of “Quark and Lepton Flavour Physics” working group in DAE-HEP 2022 symposium held at IISER Mohali.
- Organizing committee member of the Belle Analysis Workshop (BAW) series to bring together PhD students, postdoctoral fellows, and faculty working with the Belle (II) experiment.
- Member of scientific committee for the first Belle II Physics Week (6th Belle II Theory Interface Platform, B2TiP, Workshop)-2018 KEK, Tsukuba, Japan.
- Member of organizing committee for Post-CKM School-2016, TIFR, Mumbai.
- An active member of Belle II Outreach team, and managed Belle II social networking sites and Belle II webpage (2015 - till date).

Service to the Institute (IIT Hyderabad):

- Sunshine Faculty Coordinator (to help the students to make an easy and fun-filled transition from school life to the IITH) for Physics Department (Sep 2023 -)
- Member of the Institute Ranking Committee (May 2023 -)
- Radiation Professional (RP) cum Radiological Safety Officer(RSO) as a representative of the institute to the Atomic Energy Regulatory Board (AERB) (Jul 2022 -)
- Member of the Institute Safety Committee (for Radiation and Laser Safety) (Mar 2021 -)

- Member of several Departmental (at Department of Physics, IIT Hyderabad) Committees (a) Departmental Under Graduate Committee (b) Department Vision Committee (c) Department Purchase Committee (d) Department PhD Admission Committee.